SDG Missingness, Democracy, and Transparency

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Contents

1 Load Data

```
df <- read_csv("cleaned_data.csv")

sdg_cols <- df %>%
  select(starts_with("sdg")) %>%
  colnames()

length(sdg_cols)
```

```
## [1] 126
```

2 Construct SDG Missingness Dataset

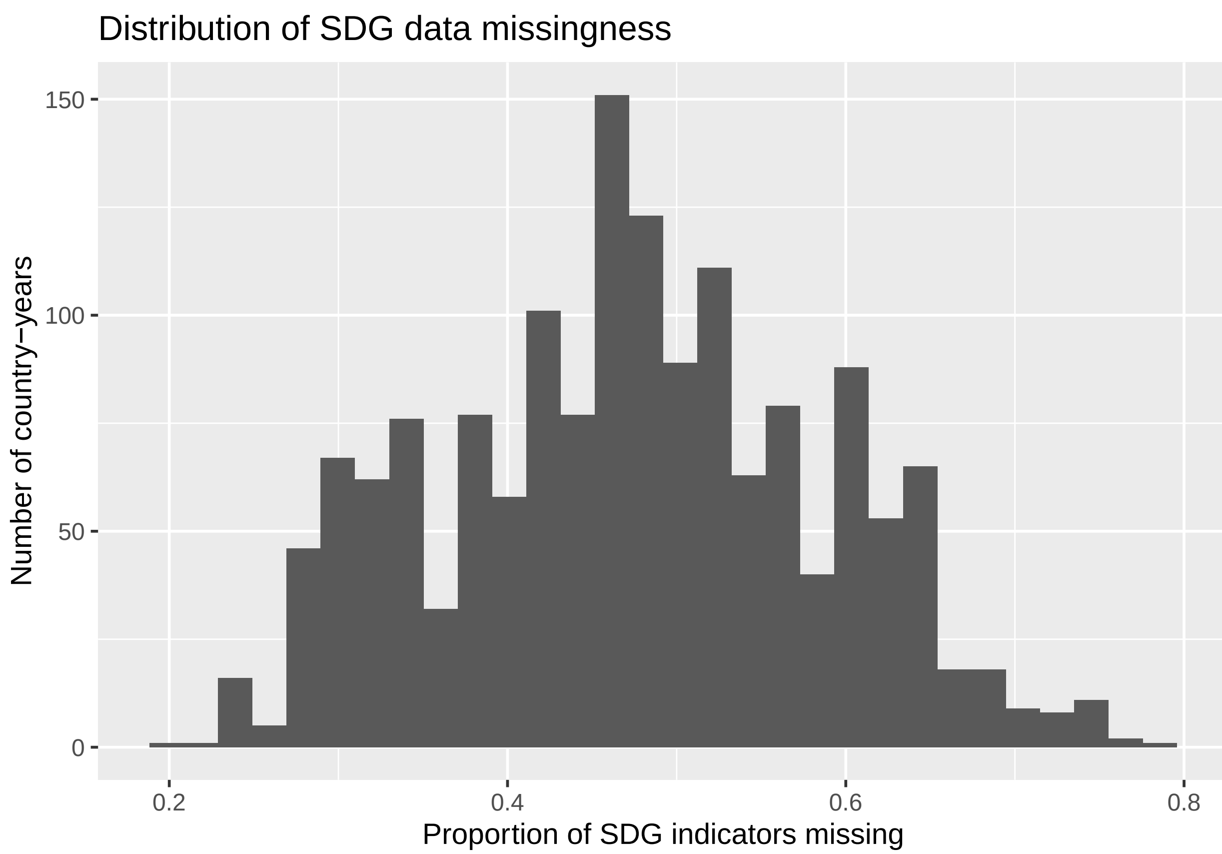
```
df <- df %>%
  mutate(
    n_sdg_indicators = length(sdg_cols),
    n_sdg_missing    = apply(select(., all_of(sdg_cols)), 1, function(x) sum(is.na(x))),
    prop_sdg_missing = n_sdg_missing / n_sdg_indicators
  )
```

2.1 Overall distribution of missingness

```
df %>%
  summarise(
    mean_missing = mean(prop_sdg_missing, na.rm = TRUE),
    sd_missing   = sd(prop_sdg_missing, na.rm = TRUE),
    min_missing  = min(prop_sdg_missing, na.rm = TRUE),
    max_missing  = max(prop_sdg_missing, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 4
##   mean_missing sd_missing min_missing max_missing
##          <dbl>      <dbl>       <dbl>       <dbl>
## 1        0.474      0.112       0.198       0.786
```

```
# Histogram
ggplot(df, aes(x = prop_sdg_missing)) +
  geom_histogram(bins = 30) +
  labs(
    x = "Proportion of SDG indicators missing",
    y = "Number of country-years",
    title = "Distribution of SDG data missingness"
  )
```



2.2 Missingness by SDG goal

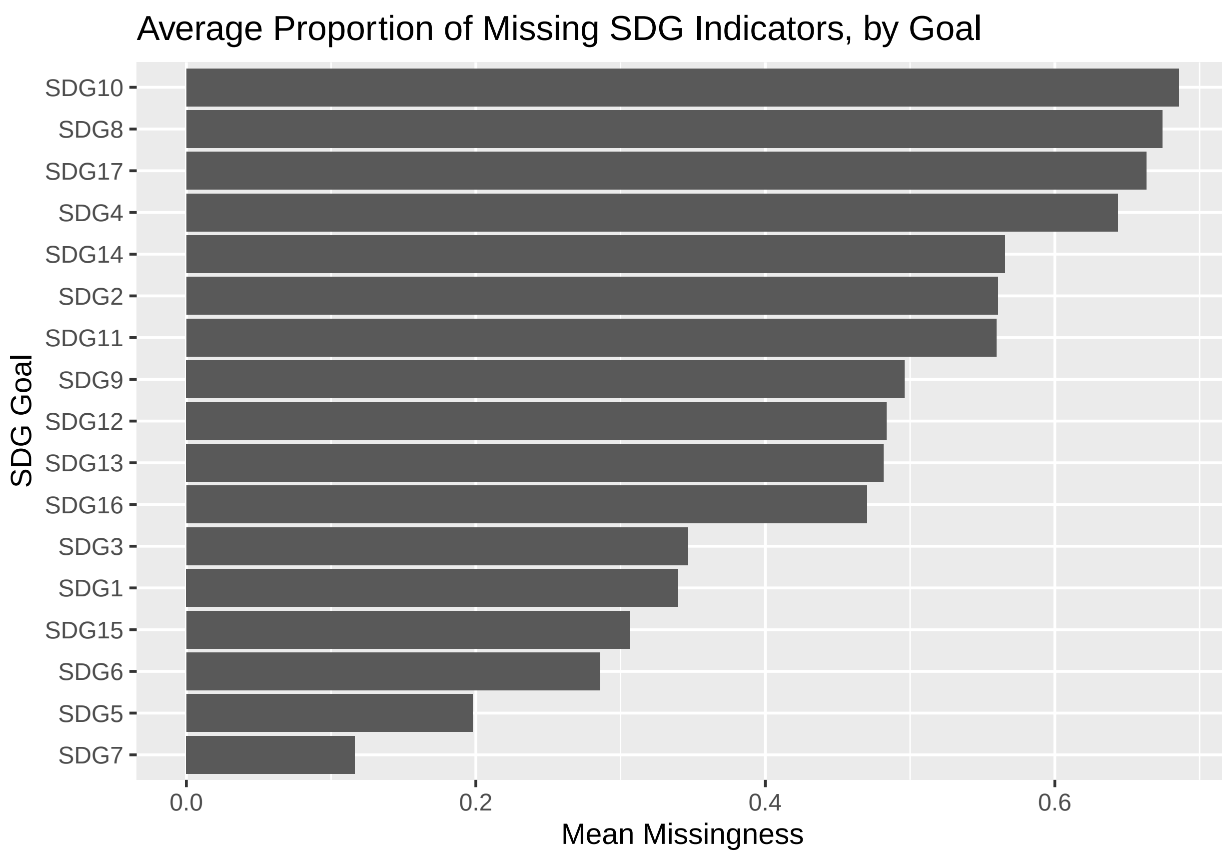
We compute proportion missing per goal (1–17) for each country-year.

```r
# Construct per-goal missingness for each country-year
goal_missing <- function(g) {
  goal_cols <- grep(paste0("^sdg", g, "_"), names(df), value = TRUE)
  df %>%
    mutate(
      goal = paste0("SDG", g),
      n_goal_ind = length(goal_cols),
      n_goal_missing = apply(select(., all_of(goal_cols)), 1, function(x) sum(is.na(x))),
      prop_missing = n_goal_missing / n_goal_ind
    ) %>%
    select(country_code, country_name, year, income_level, goal, prop_missing)
}

goal_miss_df <- bind_rows(lapply(1:17, goal_missing))

# Average missingness per goal across all country-years
goal_summary <- goal_miss_df %>%
  group_by(goal) %>%
  summarise(
    mean_prop_missing = mean(prop_missing, na.rm = TRUE),
    sd_prop_missing   = sd(prop_missing, na.rm = TRUE)
```

```r
  ) %>%
  arrange(desc(mean_prop_missing))

goal_summary
```

```
## # A tibble: 17 x 3
##    goal  mean_prop_missing sd_prop_missing
##    <chr>             <dbl>           <dbl>
##  1 SDG10             0.686           0.402
##  2 SDG8              0.674           0.143
##  3 SDG17             0.663           0.132
##  4 SDG4              0.643           0.247
##  5 SDG14             0.565           0.315
##  6 SDG2              0.561           0.186
##  7 SDG11             0.560           0.137
##  8 SDG9              0.496           0.179
##  9 SDG12             0.484           0.180
## 10 SDG13             0.482           0.127
## 11 SDG16             0.470           0.222
## 12 SDG3              0.347           0.140
## 13 SDG1              0.340           0.256
## 14 SDG15             0.307           0.124
## 15 SDG6              0.286           0.271
## 16 SDG5              0.198           0.114
## 17 SDG7              0.116           0.240
```

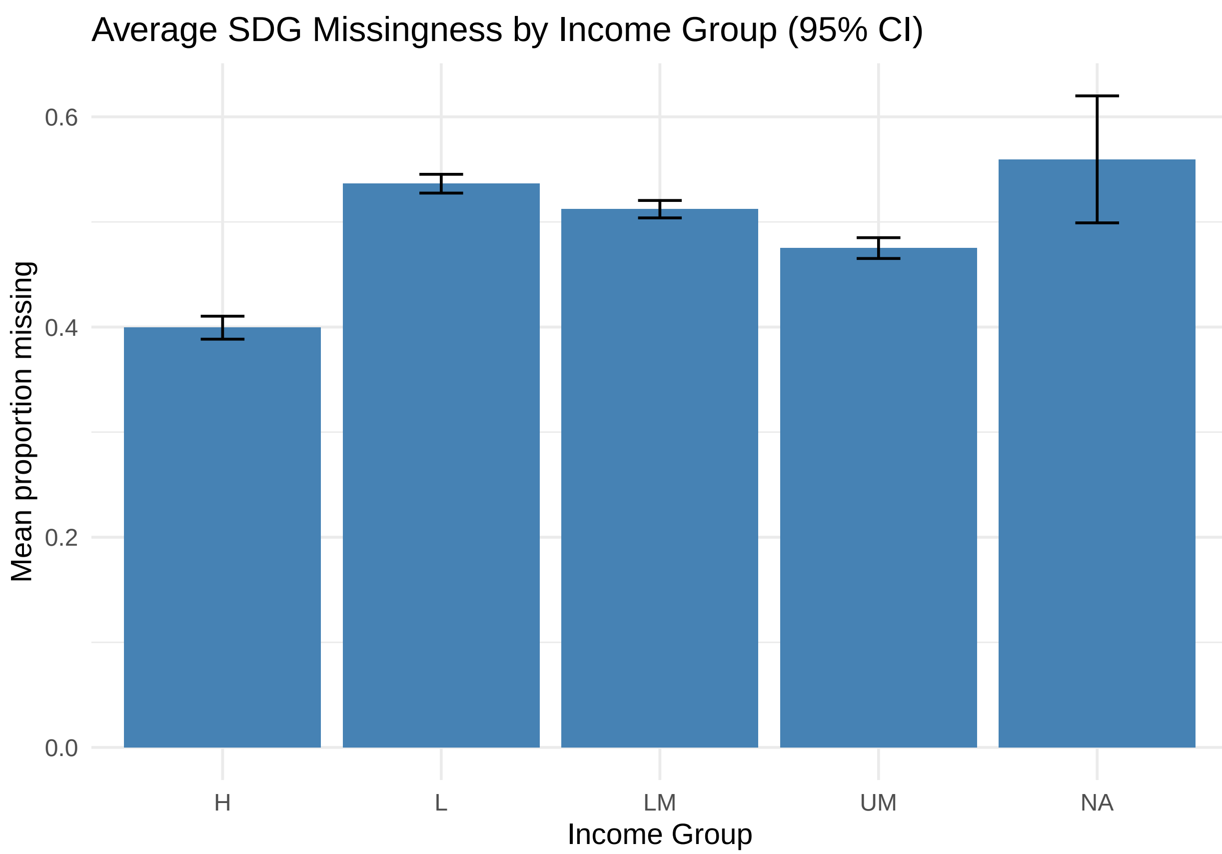
```r
#Bar plot of missingness by SDG goal

ggplot(goal_summary, aes(x = reorder(goal, mean_prop_missing), y = mean_prop_missing)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Average Proportion of Missing SDG Indicators, by Goal",
    x = "SDG Goal",
    y = "Mean Missingness"
  )
```




```r
#Pattern across country-income groups

#Overall missingness
income_missing_summary <- df %>%
  group_by(income_level) %>%
  summarise(
    mean_missing = mean(prop_sdg_missing, na.rm = TRUE),
    sd_missing   = sd(prop_sdg_missing, na.rm = TRUE),
    median_missing = median(prop_sdg_missing, na.rm = TRUE),
    n_obs = n()
  ) %>%
  arrange(desc(mean_missing))

income_missing_summary
```

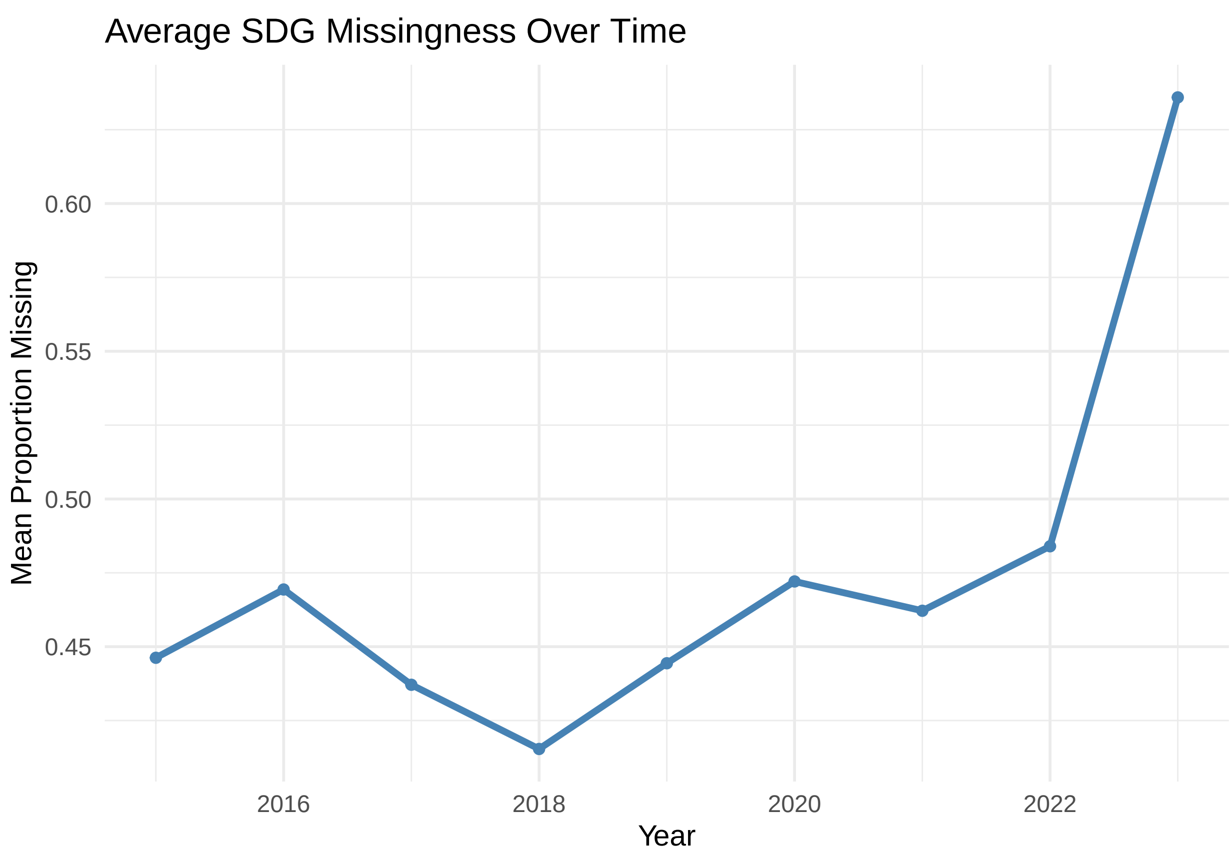
```
## # A tibble: 5 x 5
##   income_level mean_missing sd_missing median_missing n_obs
##   <chr>               <dbl>      <dbl>          <dbl> <int>
## 1 <NA>                0.560     0.0616          0.536     4
## 2 L                   0.536     0.0739          0.524   263
## 3 LM                  0.512     0.0874          0.492   427
## 4 UM                  0.475     0.102           0.468   407
## 5 H                   0.399     0.118           0.349   447
```

```r
#Bar Plot with 95% CI
income_ci <- df %>%
  group_by(income_level) %>%
  summarise(
    mean_missing = mean(prop_sdg_missing, na.rm = TRUE),
    se_missing   = sd(prop_sdg_missing, na.rm = TRUE) / sqrt(n()),
    lower_ci     = mean_missing - 1.96 * se_missing,
    upper_ci     = mean_missing + 1.96 * se_missing
  )

ggplot(income_ci, aes(x = income_level, y = mean_missing)) +
  geom_col(fill = "steelblue") +
  geom_errorbar(
    aes(ymin = lower_ci, ymax = upper_ci),
    width = 0.2
  ) +
  labs(
    title = "Average SDG Missingness by Income Group (95% CI)",
    x = "Income Group",
    y = "Mean proportion missing"
  ) +
  theme_minimal()
```



Overall Average Missingness over time

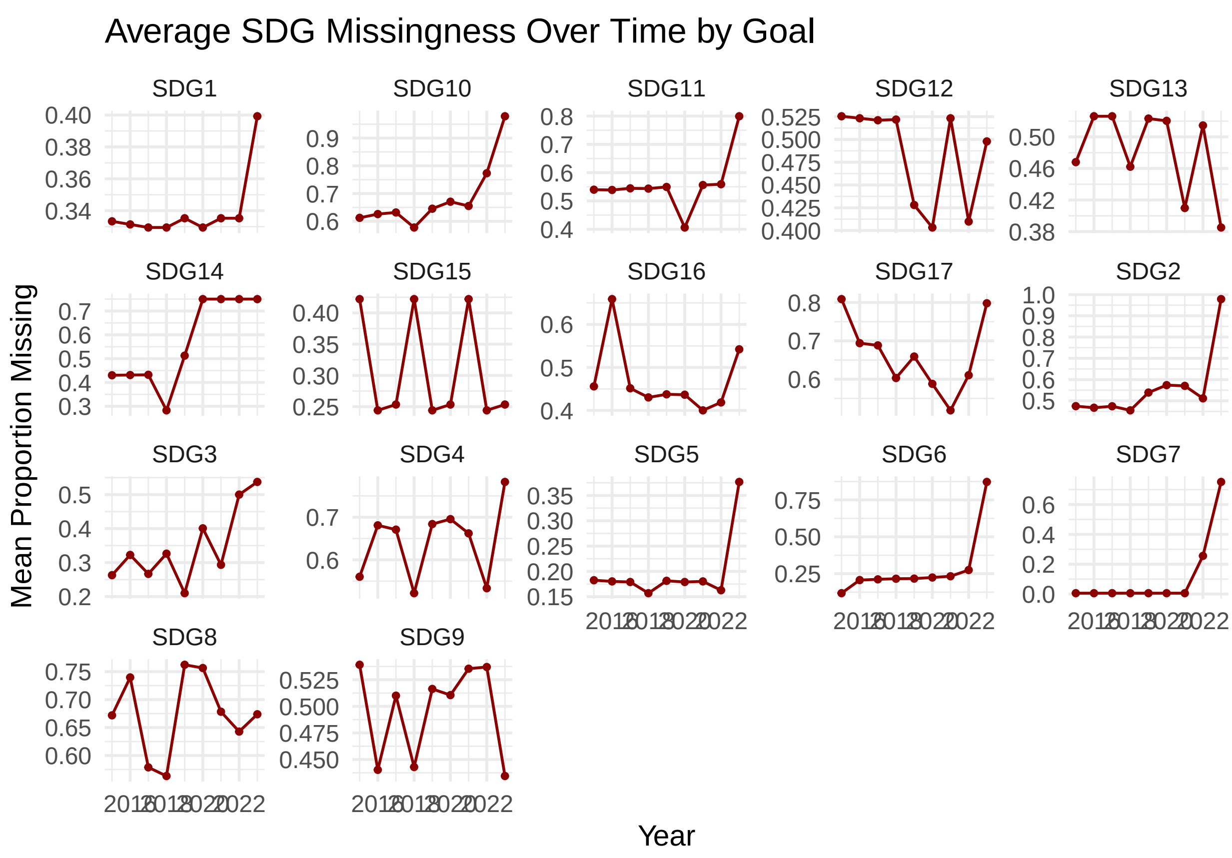
```r
overall_missing_over_time <- df %>%
  group_by(year) %>%
  summarise(
    mean_missing = mean(prop_sdg_missing, na.rm = TRUE),
    sd_missing = sd(prop_sdg_missing, na.rm = TRUE),
    n = n()
  )


ggplot(overall_missing_over_time, aes(x = year, y = mean_missing)) +
  geom_line(color = "steelblue", size = 1.2) +
  geom_point(color = "steelblue") +
  labs(
    title = "Average SDG Missingness Over Time",
    x = "Year",
    y = "Mean Proportion Missing"
  ) +
  theme_minimal()
```



#Goal-Level Average Missingness Over Time

```r
goal_missing_over_time <- goal_miss_df %>%
  group_by(goal, year) %>%
  summarise(
    mean_missing = mean(prop_missing, na.rm = TRUE),
    sd_missing = sd(prop_missing, na.rm = TRUE),
```

```
    n = n()
  ) %>%
  ungroup()

ggplot(goal_missing_over_time,
       aes(x = year, y = mean_missing)) +
  geom_line(color = "darkred") +
  geom_point(color = "darkred", size = 0.8) +
  facet_wrap(~ goal, scales = "free_y") +
  labs(
    title = "Average SDG Missingness Over Time by Goal",
    x = "Year",
    y = "Mean Proportion Missing"
  ) +
  theme_minimal()
```




```
#Patterns across per goal missingness by income group

goal_missing <- function(g) {
  goal_cols <- grep(paste0("^sdg", g, "_"), names(df), value = TRUE)

  df %>%
    mutate(
      goal = paste0("SDG", g),
      n_ind = length(goal_cols),
```

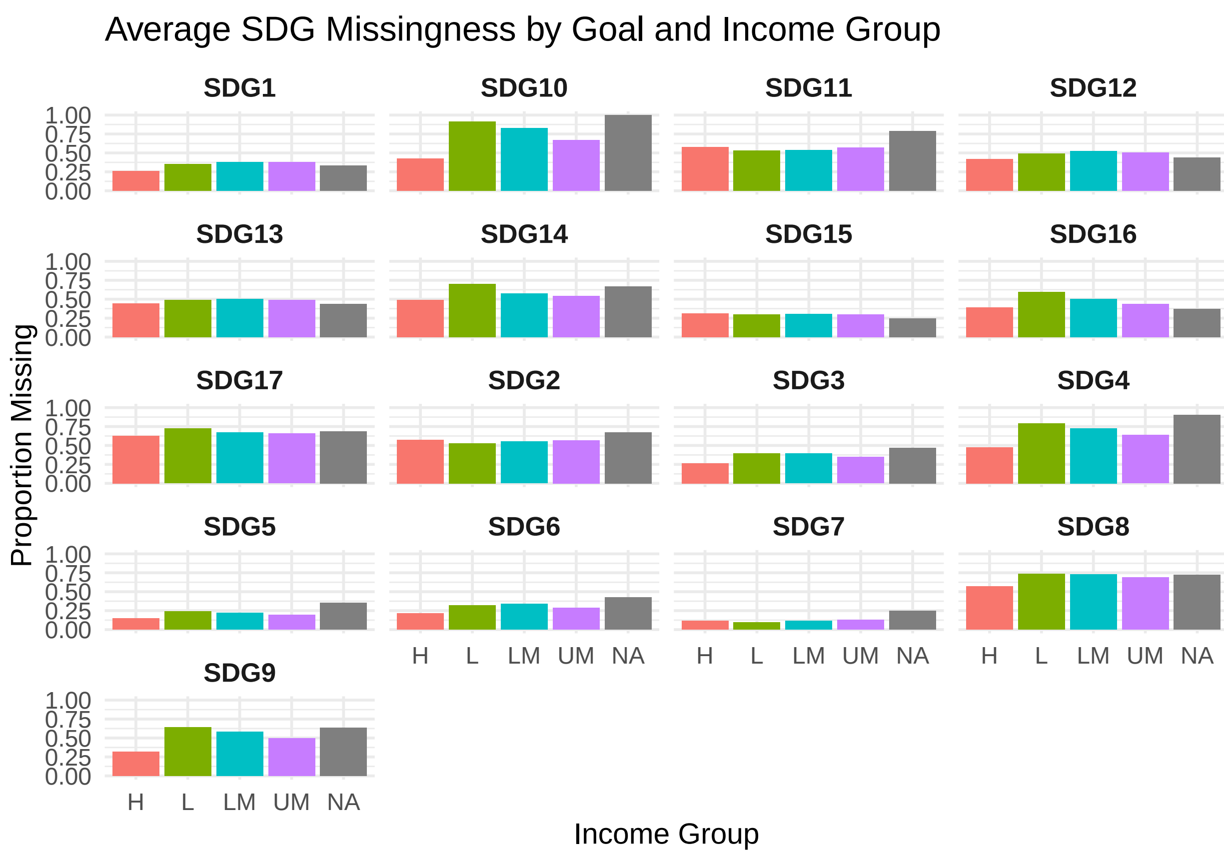
```r
      n_miss = apply(select(., all_of(goal_cols)), 1, function(x) sum(is.na(x))),
      prop_missing = n_miss / n_ind
    ) %>%
    select(
      country_code,
      country_name,
      year,
      income_level,    # <-- ADD THIS
      goal,
      prop_missing
    )
}

goal_miss_df <- dplyr::bind_rows(lapply(1:17, goal_missing))

goal_missing_over_time <- goal_miss_df %>%
  dplyr::group_by(goal, year) %>%
  dplyr::summarise(
    mean_missing = mean(prop_missing, na.rm = TRUE),
    sd_missing   = sd(prop_missing, na.rm = TRUE),
    n = dplyr::n()
  ) %>%
  dplyr::ungroup()
```

```r
goal_income_summary <- goal_miss_df %>%
  group_by(goal, income_level) %>%
  summarise(mean_missing = mean(prop_missing, na.rm = TRUE))

goal_income_summary
```

```
## # A tibble: 85 x 3
## # Groups:   goal [17]
##    goal  income_level mean_missing
##    <chr> <chr>               <dbl>
##  1 SDG1  H                   0.258
##  2 SDG1  L                   0.356
##  3 SDG1  LM                  0.379
##  4 SDG1  UM                  0.378
##  5 SDG1  <NA>                0.333
##  6 SDG10 H                   0.426
##  7 SDG10 L                   0.914
##  8 SDG10 LM                  0.829
##  9 SDG10 UM                  0.670
## 10 SDG10 <NA>                1
## # i 75 more rows
```

```r
ggplot(goal_income_summary,
       aes(x = income_level, y = mean_missing, fill = income_level)) +
  geom_col() +
  facet_wrap(~ goal, ncol = 4) +
  labs(
    title = "Average SDG Missingness by Goal and Income Group",
```

```
    x = "Income Group",
    y = "Proportion Missing"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    strip.text = element_text(size = 10, face = "bold")
  )
```

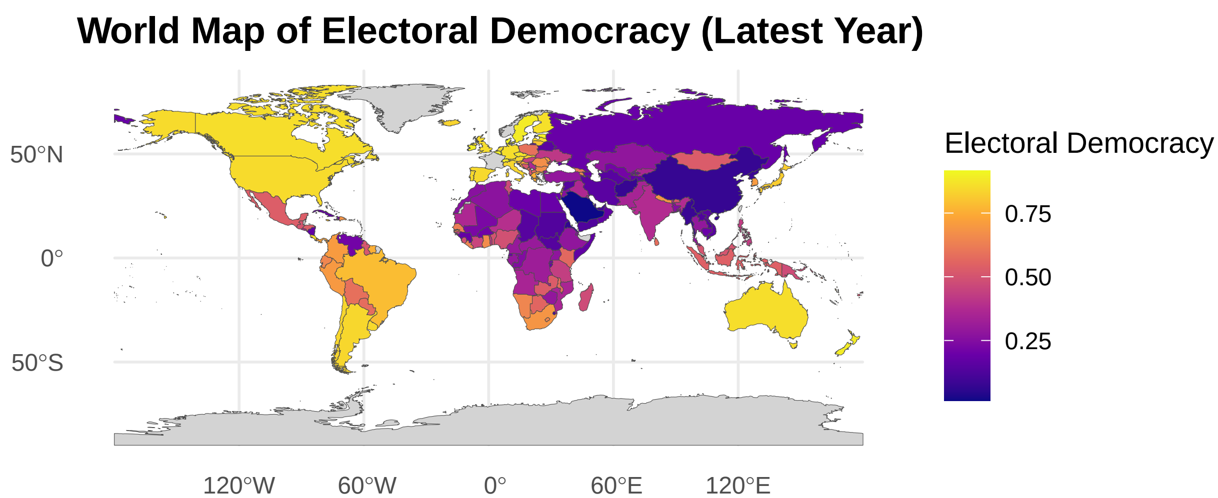


Demcoractic variables

2.3 World map of democracy levels

```
#install.packages(c("sf", "rnaturalearth", "rnaturalearthdata", "ggplot2", "dplyr"))

library(sf)
library(rnaturalearth)
library(rnaturalearthdata)

world <- ne_countries(scale = "medium", returnclass = "sf")


df_latest <- df %>%
  filter(year == max(year)) %>%
```

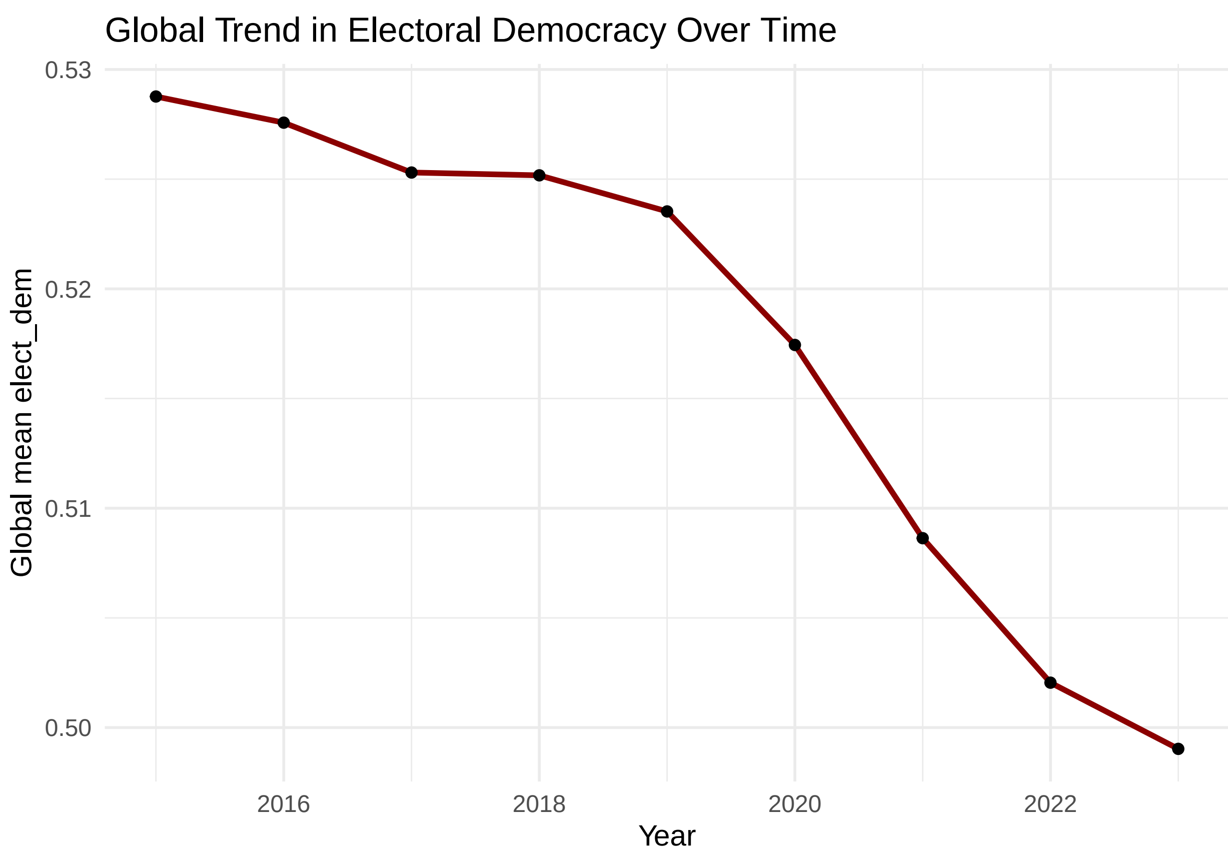
```r
  select(country_code, elect_dem)

world_dem <- world %>%
  left_join(df_latest, by = c("iso_a3" = "country_code"))

ggplot(data = world_dem) +
  geom_sf(aes(fill = elect_dem), color = "gray30", size = 0.1) +
  scale_fill_viridis_c(option = "plasma", na.value = "lightgray") +
  theme_minimal() +
  labs(
    title = "World Map of Electoral Democracy (Latest Year)",
    fill  = "Electoral Democracy"
  ) +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    legend.position = "right"
  )
```



2.4 Gobal democracy trend over time

```r
df %>%
  group_by(year) %>%
  summarise(global_mean = mean(elect_dem, na.rm = TRUE)) %>%
```

```r
ggplot(aes(x = year, y = global_mean)) +
geom_line(size = 1, color = "darkred") +
geom_point() +
theme_minimal() +
labs(
  title = "Global Trend in Electoral Democracy Over Time",
  x = "Year",
  y = "Global mean elect_dem"
)
```



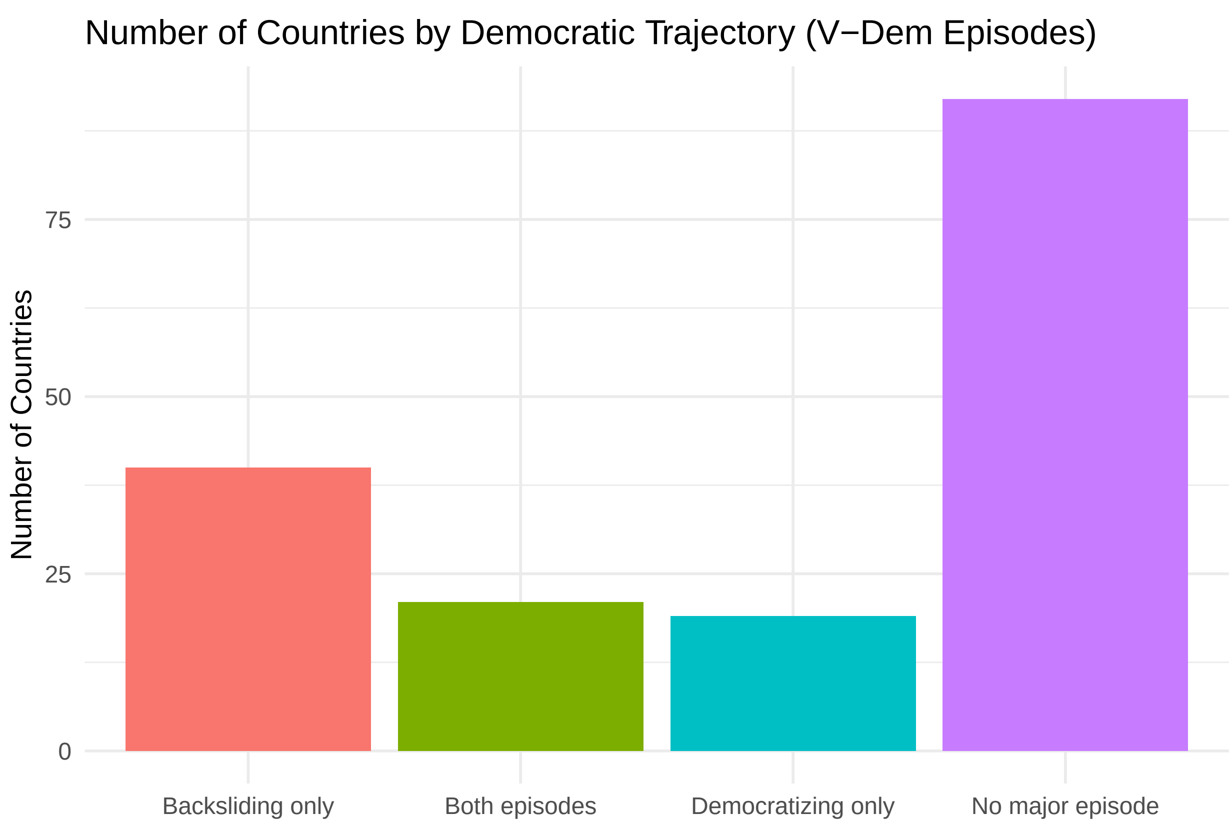
2.5 How many countries experienced democratization / backsliding?

```r
dem_groups <- df %>%
  group_by(country_code, country_name) %>%
  summarise(
    ever_dem = any(dem_ep == 1, na.rm = TRUE),
    ever_aut = any(aut_ep == 1, na.rm = TRUE)
  ) %>%
  mutate(
    trajectory = case_when(
      ever_aut & !ever_dem ~ "Backsliding only",
      ever_dem & !ever_aut ~ "Democratizing only",
      ever_dem & ever_aut  ~ "Both episodes",
```

```
      TRUE                    ~ "No major episode"
    )
  )


table(dem_groups$trajectory)
```

```
##
##    Backsliding only      Both episodes Democratizing only   No major episode
##                  40                 21                 19                 92
```

```
ggplot(dem_groups, aes(x = trajectory, fill = trajectory)) +
  geom_bar() +
  labs(
    title = "Number of Countries by Democratic Trajectory (V-Dem Episodes)",
    x = "",
    y = "Number of Countries"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



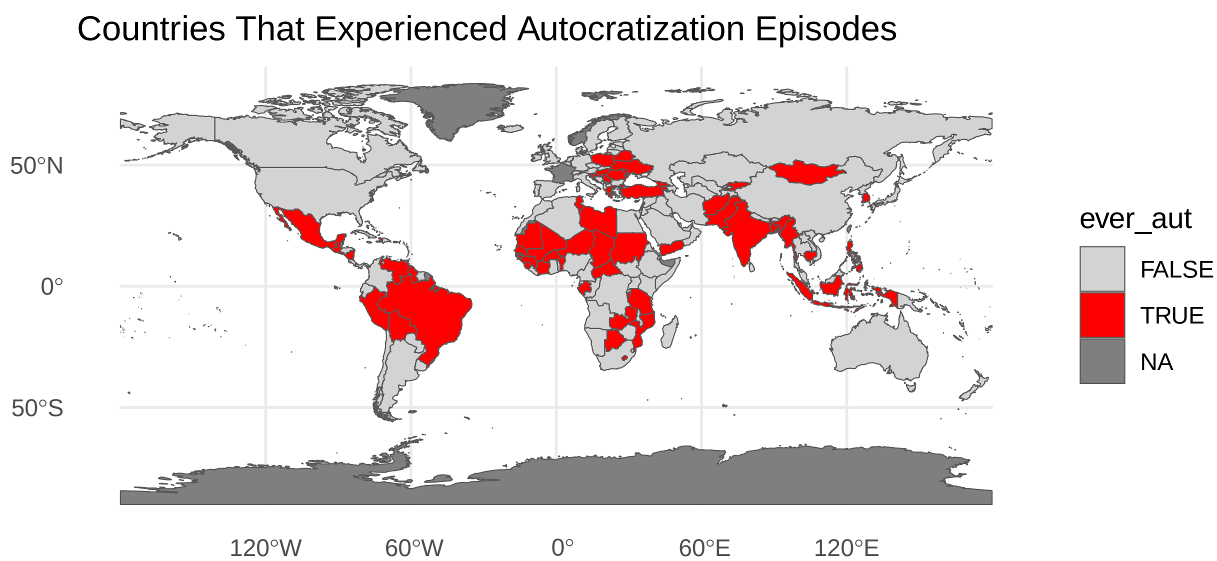

```
only_backsliding <- dem_groups %>%
  filter(trajectory == "Backsliding only") %>%
  arrange(country_name)
```

```r
only_backsliding
```

```
## # A tibble: 40 x 5
## # Groups:   country_code [40]
##    country_code country_name ever_dem ever_aut trajectory
##    <chr>        <chr>        <lgl>    <lgl>    <chr>
##  1 AFG          Afghanistan  FALSE    TRUE     Backsliding only
##  2 BGD          Bangladesh   FALSE    TRUE     Backsliding only
##  3 BLR          Belarus      FALSE    TRUE     Backsliding only
##  4 BWA          Botswana     FALSE    TRUE     Backsliding only
##  5 BDI          Burundi      FALSE    TRUE     Backsliding only
##  6 KHM          Cambodia     FALSE    TRUE     Backsliding only
##  7 TCD          Chad         FALSE    TRUE     Backsliding only
##  8 COM          Comoros      FALSE    TRUE     Backsliding only
##  9 HRV          Croatia      FALSE    TRUE     Backsliding only
## 10 SLV          El Salvador  FALSE    TRUE     Backsliding only
## # i 30 more rows
```

2.6 Map autocratization episodes

```r
df_auto <- df %>%
  group_by(country_code) %>%
  summarise(ever_aut = any(aut_ep == 1, na.rm = TRUE))

world_auto <- world %>%
  left_join(df_auto, by = c("iso_a3" = "country_code"))

ggplot(world_auto) +
  geom_sf(aes(fill = ever_aut)) +
  scale_fill_manual(values = c("lightgray", "red")) +
  theme_minimal() +
  labs(title = "Countries That Experienced Autocratization Episodes")
```



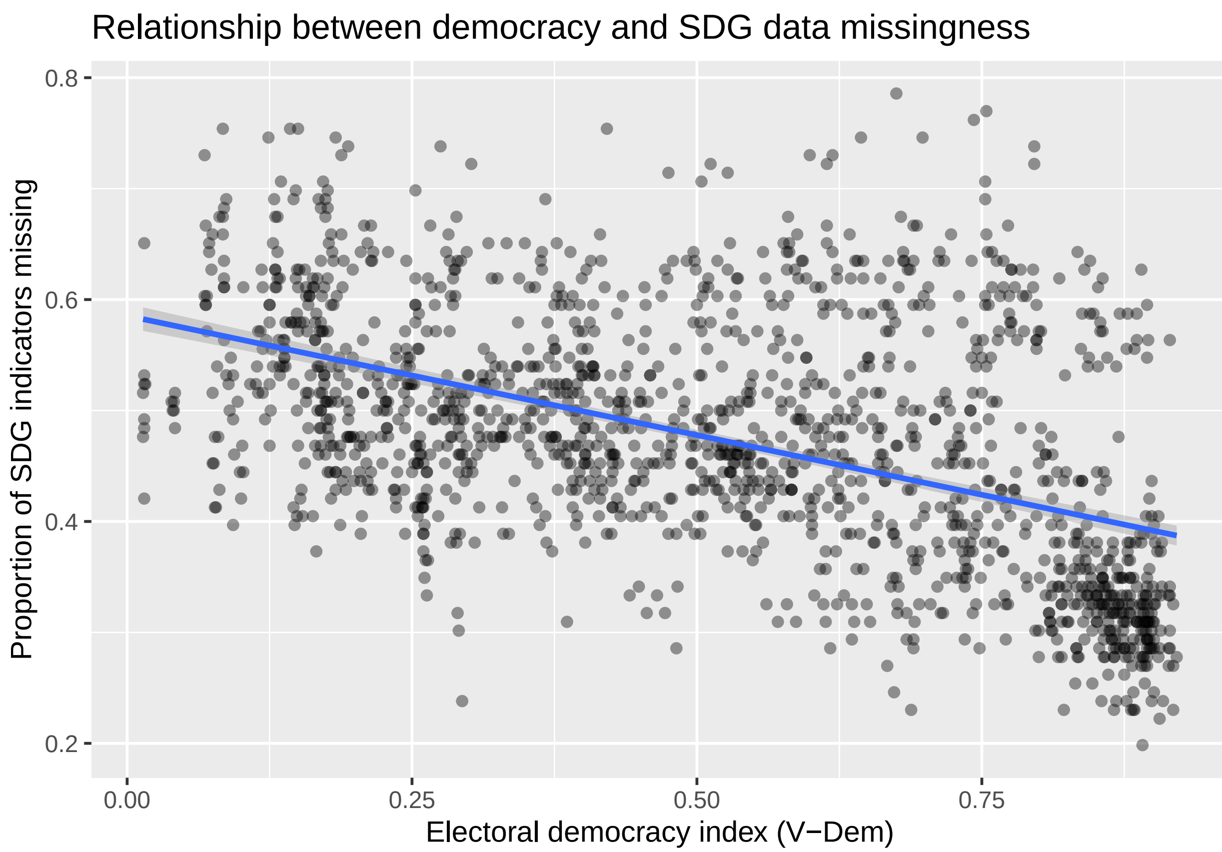
3 Missingness vs Democracy

3.1 Correlation and scatter plots: democracy vs missingness

```
df %>%
  select(prop_sdg_missing, elect_dem, lib_dem, part_dem, egal_dem, delib_dem) %>%
  cor(use = "pairwise.complete.obs")
```

```
##                  prop_sdg_missing  elect_dem    lib_dem   part_dem   egal_dem
## prop_sdg_missing        1.0000000 -0.4970093 -0.5162339 -0.5477242 -0.5270796
## elect_dem              -0.4970093  1.0000000  0.9784190  0.9710565  0.9473120
## lib_dem                -0.5162339  0.9784190  1.0000000  0.9666637  0.9708041
## part_dem               -0.5477242  0.9710565  0.9666637  1.0000000  0.9461906
## egal_dem               -0.5270796  0.9473120  0.9708041  0.9461906  1.0000000
## delib_dem              -0.5100436  0.9674004  0.9785391  0.9567687  0.9626232
##                   delib_dem
## prop_sdg_missing -0.5100436
## elect_dem         0.9674004
## lib_dem           0.9785391
## part_dem          0.9567687
## egal_dem          0.9626232
## delib_dem         1.0000000
```

```r
ggplot(df, aes(x = elect_dem, y = prop_sdg_missing)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "Electoral democracy index (V-Dem)",
    y = "Proportion of SDG indicators missing",
    title = "Relationship between democracy and SDG data missingness"
  )
```



3.2 Other plots for democratic variables

4 6. Two-Way Fixed Effects Models with staggered treatment

Question: Does entering an autocratization episode increase SDG data missingness?

4.1 Define treatment timing (staggered adoption)

We use aut_ep_start_yr as the group (treatment year). For countries that never backslide, this is NA.

```r
treat_df <- df %>%
  group_by(country_code) %>%
  summarise(
```

```r
    G_aut = suppressWarnings(min(aut_ep_start_yr[aut_ep == 1], na.rm = TRUE)),
    .groups = "drop"
  ) %>%
  mutate(
    G_aut = ifelse(is.infinite(G_aut), NA, G_aut)   # if no episode, set to NA
  )

df2 <- df %>%
  left_join(treat_df, by = "country_code") %>%
  mutate(
    # TWFE-like treatment indicator: treated once year >= G_aut
    D_aut = ifelse(!is.na(G_aut) & year >= G_aut, 1, 0)
  )
```

4.2 TFWE Model

Outcome: prop_sdg_missing Treatment: D_aut Fixed effects: country and year Errors: clustered by country

```r
twfe_model <- feols(
  prop_sdg_missing ~ D_aut | country_code + year,
  cluster = ~ country_code,
  data = df2
)

summary(twfe_model)
```

```
## OLS estimation, Dep. Var.: prop_sdg_missing
## Observations: 1,548
## Fixed-effects: country_code: 172,  year: 9
## Standard-errors: Clustered (country_code)
##       Estimate Std. Error t value Pr(>|t|)
## D_aut 0.000909   0.003677 0.24709  0.80513
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.025202     Adj. R2: 0.942259
##                  Within R2: 4.396e-5
```

```r
twfe_model2 <- feols(
  prop_sdg_missing ~ D_aut + log_gdppc + log_pop | country_code + year,
  cluster = ~ country_code,
  data = df2
)
summary(twfe_model2)
```

```
## OLS estimation, Dep. Var.: prop_sdg_missing
## Observations: 1,342
## Fixed-effects: country_code: 168,  year: 8
## Standard-errors: Clustered (country_code)
##            Estimate Std. Error   t value Pr(>|t|)
## D_aut      0.002025   0.003351  0.604460  0.54636
```

```
## log_gdppc -0.013658   0.005649 -2.418027  0.01668 *
## log_pop    0.004645   0.028150  0.165013  0.86913
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.020334     Adj. R2: 0.950463
##                  Within R2: 0.005992
```

Coefficient on D_aut = average change in missingness after a country enters autocratization, relative to never-treated and not-yet-treated country-years, controlling for country FE and year FE.

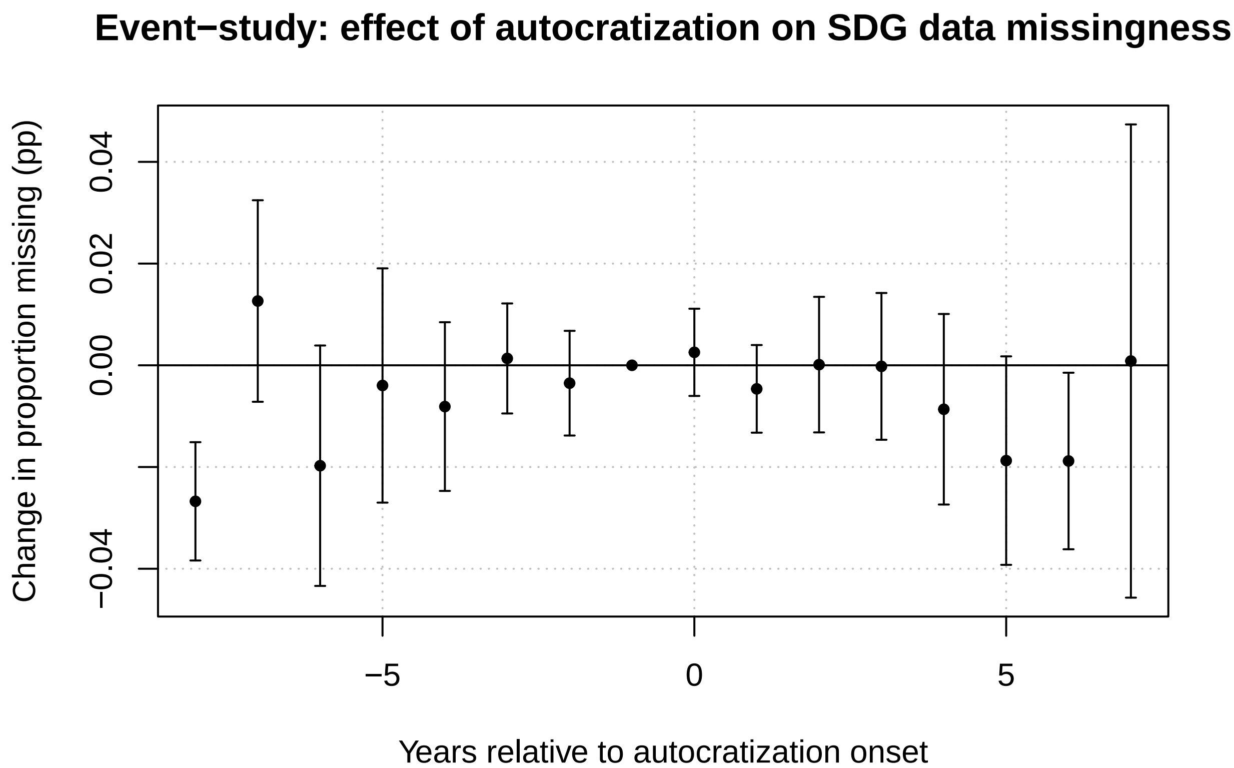
5 Event-study (staggered DID)

Use sunab() in fixest with group = treatment year (G_aut), time = calendar year (year).

```
es_model <- feols(
  prop_sdg_missing ~ sunab(G_aut, year) | country_code + year,
  cluster = ~ country_code,
  data = df2
)

summary(es_model)
```

```
## OLS estimation, Dep. Var.: prop_sdg_missing
## Observations: 548
## Fixed-effects: country_code: 61,  year: 9
## Standard-errors: Clustered (country_code)
##           Estimate Std. Error   t value   Pr(>|t|)
## year::-8 -0.026749   0.005815 -4.599805 2.2362e-05 ***
## year::-7  0.012640   0.009904  1.276231 2.0679e-01
## year::-6 -0.019743   0.011816 -1.670933 9.9944e-02 .
## year::-5 -0.003968   0.011513 -0.344670 7.3155e-01
## year::-4 -0.008116   0.008290 -0.979025 3.3150e-01
## year::-3  0.001347   0.005407  0.249145 8.0410e-01
## year::-2 -0.003516   0.005144 -0.683570 4.9688e-01
## year::0   0.002550   0.004286  0.595051 5.5405e-01
## year::1  -0.004632   0.004307 -1.075374 2.8651e-01
## year::2   0.000133   0.006662  0.019927 9.8417e-01
## year::3  -0.000210   0.007214 -0.029105 9.7688e-01
## year::4  -0.008645   0.009365 -0.923042 3.5968e-01
## year::5  -0.018739   0.010247 -1.828682 7.2419e-02 .
## year::6  -0.018812   0.008677 -2.168125 3.4126e-02 *
## year::7   0.000840   0.023257  0.036111 9.7131e-01
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.022428     Adj. R2: 0.934767
##                  Within R2: 0.132045
```

```
# Plot dynamic effects
iplot(es_model,
      main = "Event-study: effect of autocratization on SDG data missingness",
      xlab = "Years relative to autocratization onset",
      ylab = "Change in proportion missing (pp)")
```




```r
etable(es_model, se = "cluster")
```

```
##                                 es_model
## Dependent Var.:        prop_sdg_missing
##
## year = -8        -0.0268*** (0.0058)
## year = -7            0.0126 (0.0099)
## year = -6          -0.0197. (0.0118)
## year = -5           -0.0040 (0.0115)
## year = -4           -0.0081 (0.0083)
## year = -3            0.0013 (0.0054)
## year = -2           -0.0035 (0.0051)
## year = 0             0.0025 (0.0043)
## year = 1            -0.0046 (0.0043)
## year = 2             0.0001 (0.0067)
## year = 3            -0.0002 (0.0072)
## year = 4            -0.0086 (0.0094)
## year = 5           -0.0187. (0.0103)
## year = 6           -0.0188* (0.0087)
## year = 7             0.0008 (0.0233)
## Fixed-Effects:   -------------------
## country_code                     Yes
## year                             Yes
## _______________  ___________________
## S.E.: Clustered    by: country_code
```

```
## Observations                        548
## R2                              0.95051
## Within R2                       0.13205
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
df_aut <- df %>%
  group_by(country_code, country_name) %>%
  mutate(
    ever_aut = any(aut_ep == 1, na.rm = TRUE)
  ) %>%
  ungroup()

focus_country <- "Turkey"   # <-- change to whatever country you want

treat_year <- df_aut %>%
  filter(country_name == focus_country, aut_ep == 1) %>%
  summarise(first_year = min(year, na.rm = TRUE)) %>%
  pull(first_year)

#Keep only that country + never-autocratized countries
data_focus <- df_aut %>%
  filter(country_name == focus_country | ever_aut == FALSE)

# create group label like Harold's evertreated factor
data_focus <- data_focus %>%
  mutate(group = ifelse(country_name == focus_country,
                        focus_country,
                        "never autocratized"))

#parallel trend averages
data_focus_bygrp <- data_focus %>%
  group_by(year, group) %>%
  summarise(
    mean_missing = mean(prop_sdg_missing, na.rm = TRUE),
    .groups = "drop"
  )


#plot
data_focus_bygrp %>%
  ggplot(aes(x = year,
             y = mean_missing,
             group = group,
             color = group)) +
  geom_line() +
  geom_vline(xintercept = treat_year - 1,   # year before autocratization starts
             linetype = "dashed",
             color = "grey20",
             size = 0.8) +
  theme_minimal() +
  ylab("Mean SDG data missingness") +
  xlab("Year") +
  guides(color = guide_legend(title = "")) +
```

```
theme(legend.position = "bottom")
```